Ajay Kumar Dwivedi

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Profile

Database Reliability and Site Reliability Engineer with 15+ years of experience keeping SQL Server and PostgreSQL platforms highly available and automating infrastructure through DevOps CI/CD pipelines, using scripting tools such as PowerShell, Python, Ansible, GitHub Actions, and containers. I build monitoring and alerting systems with Python, PowerShell, Flask, Prometheus, and Grafana for both on-premises and cloud platforms (AWS and Azure).

Core Competencies

- **Databases:** SQL Server, PostgreSQL, MongoDB, T-SQL, Performance Tuning, Query Optimization, Data Migration and Upgrades, NoSQL, Performance Troubleshooting
- **SQL Server DBA:** Always On Availability Groups, Failover Clustering, Log Shipping, Database Mirroring, Transactional Replication, Change Data Capture (CDC), Index Maintenance, Table Partitioning, ScriptOut and Disaster Recovery, Failover Cluster Instances, Amazon RDS for SQL Server, SQL Server on Azure VMs
- **PostgreSQL DBA:** High Availability (Patroni), Amazon RDS and Aurora PostgreSQL, Database Migration (CentOS to RHEL), Backup and Alerting, Ansible and Terraform Automation
- **Reliability and Incidents:** Database Reliability Engineering, High Availability and DR, Reliability Engineering, Incident Management, Root Cause Analysis, On-Call, Capacity Planning, Customer-Facing, Vulnerability Assessment
- **Cloud and Containers:** AWS, Azure, Cloud Infrastructure, EC2, RDS, Kubernetes, Distributed Systems, Virtualization, CloudWatch
- **Automation and Scripting:** Infrastructure as Code, Automation, Terraform, Ansible, CI/CD, GitHub Actions, PowerShell, Python, Shell Scripting, Open-Source Development, Cost Optimization, Patch Management, Deployment Automation, Resource Management, SolarWinds DPA
- **Monitoring and AI:** Monitoring and Observability, Prometheus, Grafana, Datadog, AI Agents (LangChain, Ollama), SQL Server Profiler

Skills

- **High Availability and Disaster Recovery:** Clustering, Availability Groups, Mirroring, Log Shipping, Replication, Disk Stripping, RAID, Multi-Subnet Setup
- **Databases:** SQL Server, PostgreSQL, SQLite, RDS, Azure SQL, Managed Instance, Aurora
- **Cloud Platforms:** AWS, Azure
- **Programming Languages:** SQL, PowerShell, Python, JavaScript, HTML, CSS, Shell Scripting
- **Tools and Frameworks:** Git, Linux, Ansible, Flask, GitHub Actions, VS Code, Grafana, Prometheus, Containers, Docker, Ubuntu, RHEL, PowerShell DSC, Infrastructure as Code (IaC)
- **Languages:** English (proficient), Hindi (native)

Experience

Tessell *February 2026 - September 2026*
Principal Solution Architect - SQL Server

- Built DBaaS capabilities for deploying SQL Server on AWS EC2 and Azure VMs, enabling self-service concurrent deployments
- Wrote Python-based SAGA workflows on Kubernetes and a dockerized environment to build Tes-

sell's Database as a Service (DBaaS) platform, a distributed systems architecture with Docker and Kubernetes as the control plane and AWS, Azure, and GCP as the data plane, reducing provisioning time to 20 minutes
- Designed CentOS-to-RHEL migration automation for PostgreSQL using Python and Ansible, migrating 180+ database clusters with 15 minutes of downtime each
- Added monitoring for data and log file growth using SQLExporter, Prometheus, and VictoriaMetrics
- Added enhancements to support Change Data Capture (CDC) and transactional replication
- Used root cause analysis to debug production issues and implement fixes for customers during 24x7 on-call support
- Provided technical leadership on PostgreSQL architecture for Tessell's DBaaS SaaS platform, covering provisioning, high availability, and day 2 operations, and mentoring engineers

AngelOne *February 2022 - February 2026*
Senior Site Reliability Engineer (DevOps)

- Implemented single-click automation for Transparent Data Encryption (TDE) certificate rotation, helped customers implement Always Encrypted, upgraded SSL and TLS drivers, and built a credential manager for identity and access management, securely storing and retrieving credentials for DBA automation
- Automated common database team activities, such as SQL Server installation, configuration, monitoring, and alerting, using PowerShell, Flask, Python, T-SQL, Grafana, and PagerDuty for on-premises and cloud Managed Instances
- Tuned SQL Server queries by analyzing execution plans to fix forced serialization, scalar UDF inlining, lazy spools, cardinality misestimates, wrong join types, and residual I/O performance issues, performance bottlenecks, and deadlocks
- Automated AWS database services across IaaS and PaaS, including RDS and Aurora PostgreSQL, deployment and maintenance using Terraform
- Created a GitHub repo-based approach to baselining all Windows and Linux servers, simplifying database monitoring with Prometheus and Grafana
- Built the SQLMonitor tool with dashboards that supported capacity planning and estimated storage needs for the next couple of years based on past growth
- Automated disk, connectivity, Patroni health, and backup alerts for PostgreSQL servers using Python
- Automated multi-datacenter HA and standalone PostgreSQL setup using Ansible, saving 80 hours per week of SRE effort with a single-click GitHub Actions workflow for cluster deployment and configuration
- Provided production support for PostgreSQL databases on a 24x7 on-call rotation, meeting SLA commitments for availability and incident response
- Owned business continuity for PostgreSQL clusters, maintaining and testing failover and recovery procedures
- Led rolling upgrades and database upgrades of PostgreSQL clusters with minimal downtime, using Patroni-managed failover
- Designed read replicas for PostgreSQL using streaming replication, forming replica sets that route read operations away from the primary
- Designed sharding, table partitioning, and data lifecycle management for large PostgreSQL datasets, including time-series tables with retention, archival, and purge policies
- Used explain plans (EXPLAIN and EXPLAIN ANALYZE) to troubleshoot slow query performance and fix plan regressions in PostgreSQL
- Established database best practices and standards for PostgreSQL configuration, backups, and security, and delivered internal training on them
- Supported Redis caching for PostgreSQL-backed applications, troubleshooting cache and database interaction issues
- Performed SQL Server administration across production instances, including patching, upgrades, and configuration
- Ran database health checks and statistics maintenance on SQL Server and PostgreSQL instances,

reporting findings and remediating issues
- Implemented database security controls for SQL Server instances, including encryption, access hardening, and auditing of privileged access to meet regulatory compliance requirements
- Handled request management for database access, change, and provisioning requests within the team's incident and change workflow
- Developed runbooks, knowledge base articles, and documentation for database operations, enabling on-call engineers to resolve incidents

Arcesium *May 2020 - February 2022*
Database Reliability Engineer

- Improved database reliability by automating ScriptOut of entire SQL Server instances for disaster recovery
- Applied reliability engineering as a Database Reliability Engineer, upgrading 30 critical SQL clusters and AGs in 12 hours to avoid a week-long downtime
- Built single-click audit infrastructure for databases so application teams can set up auditing and generate reports on table usage, data modifications, security events, and data changes
- Resolved hundreds of incidents by moving 38 TB of AG databases with PowerShell
- Built automation to prevent common issues such as tempdbSaver and LogSaver

TiVo *February 2018 - May 2020*
Database Administrator

- Built a resilient backup strategy with Ola Hallengren's maintenance solution, automated scheduled refreshes of non-prod environments, set up log shipping from one prod server to another, and enabled single-click full recovery of a SQL Server instance or database
- Saved 40+ hours by automating SQL Server deployment and configuration using PowerShell
- Reduced transactional replication issues by 80% using latency monitoring infrastructure
- Implemented monitoring and alerting with Datadog
- Implemented a change management process using SQL Database Projects, SQL DACPAC, and GitHub Actions
- Reduced performance incidents by 80% by building Delta Index Defragmentation automation for replicated and HADR databases

United Health Group *October 2016 - February 2018*
Senior SQL DBA

- Saved $1 million per year through DB space capacity automation
- Handled incident management for production issues, debugging customer issues and implementing fixes during 24x7 on-call support
- Practiced stakeholder management by directly interacting with customers to fix critical customer-facing performance and deployment issues
- Improved performance issues by 50% and reduced VLDB maintenance time

Ericsson Global *June 2015 - October 2016*
Technical Lead

- Achieved a 3x performance improvement through advanced query optimization, such as recursive CTEs and table partitioning
- Mentored 18+ developers in following best practices and guidelines

TATA Consultancy Services *March 2011 - June 2015*
SQL DBA Developer

- Saved 20+ hours per week by automating core DBA tasks using dynamic SQL
- Moved 250+ large tables, each with billions of records, using SSIS in 45 hours

- Built data warehouse ETL pipelines using Pentaho and Kettle for high-volume liquor retail stores

Projects

SQLMonitor *January 2021 - Present*

Open-source, enterprise-grade SQL Server monitoring tool (GitHub)

- Built using T-SQL, PowerShell, Python, and Grafana
- Monitoring 500+ SQL Servers across environments
- Sends observability alerts to Slack, PagerDuty, and email
- Built-in auto-actions to clear and resolve issues

SQLDBATools *July 2021 - February 2023*

PowerShell module with cmdlets for SQL Server DBA activities (GitHub)

- Covers server inventory, disk and space estimation, backup history, permission audits, and maintenance job setup, reducing manual DBA effort by 60% across 120+ servers
- Built a monitoring and alerting system with Grafana dashboards, auto-clearing alerts, email notifications, and alert suppression
- Published to PSGallery for installation with `Install-Module`

AI Agent *October 2025 - Present*

LLM-powered SQL agent that answers natural-language performance questions about SQL Server instances monitored by SQLMonitor (GitHub)

- Built using Python, LangChain, Ollama, Streamlit, and SQLMonitor
- Grounds every answer in SQL queries against SQLMonitor data, shows the query in each explanation, and includes prompt guardrails against write statements, enabling application teams to troubleshoot performance problems themselves

HadrSync *January 2021 - Present*

PowerShell module that synchronizes all objects across replicas in Availability Groups (GitHub)

- Ensures zero sync issues during failovers and DR events
- Detects and synchronizes delta changes, such as changes to passwords, definitions, and more

Certifications

- Microsoft Certified Professional - Database Querying (October 2013)
- Microsoft Certified Professional - Database Administration (December 2013)
- BrentOzarUnlimited Mastering SQL Server (November 2022)
- SQLSkills Blackbelt Professional (June 2023)

Education

Bachelor of Engineering in Electronics and Communication Engineering *July 2006 - June 2010*

RustamJi Institute of Technology, Gwalior, Madhya Pradesh

Awards and Honors

- **Tech Titan**, AngelOne for 10x performance optimization on production SQL Server (2022)
- **Sapphire**, United Health Group for innovation and automation (2017)

Availability

Notice Period: 7 Days, Open to Relocation, Open to remote opportunities